

# Collaborative AI Scaffolding for Structured Drawing in Dementia Care: A Feasibility Study

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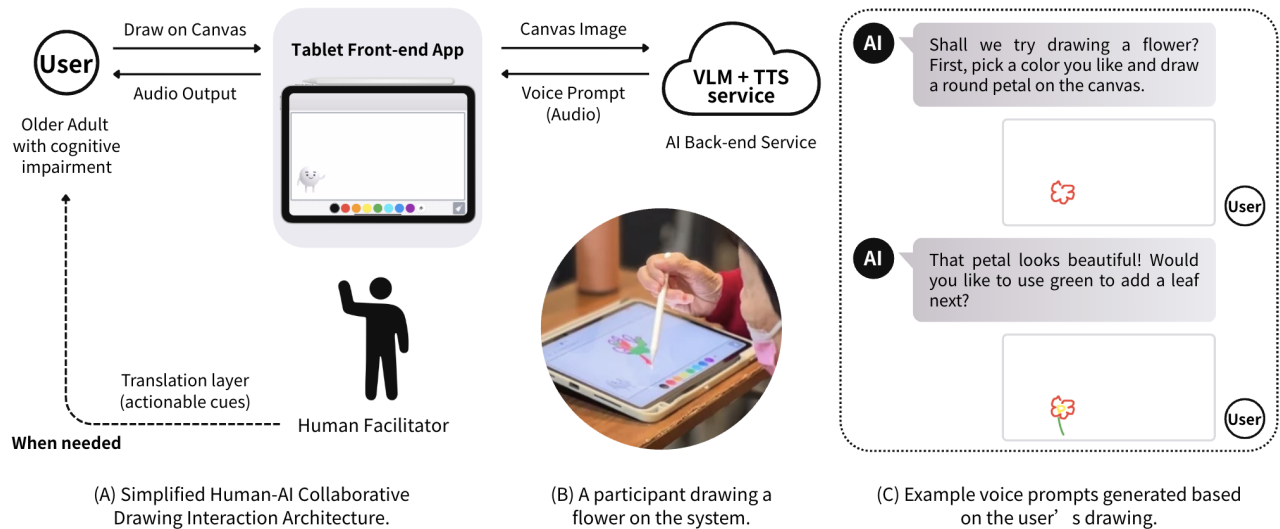


Figure 1: Human-in-the-Loop (HITL) collaborative AI scaffolding for structured drawing in dementia care.

## Abstract

We present a tablet-based AI drawing tool for older adults with dementia in structured creative activities. The system functions as a human-in-the-loop (HITL) facilitator, providing collaborative scaffolding by integrating a vision-language model with localized text-to-speech to provide short, stepwise voice cues through actionable micro-prompts, paired with a low-clutter interface and an animated companion. To avoid overlapping speech, the backend defers new triggers during audio playback and batches pending strokes into a single post-playback update. In a single-group feasibility study in a day-care center ( $n=6$ ; CDR 1–4), 5/6 participants completed all four target elements with AI plus optional human facilitation, while only 2/6 did so without human help. Video analysis reveals a semantics-to-action gap: participants followed concrete cues (shapes, color keywords, deictic actions) more reliably than abstract part labels, and facilitators often translated prompts into executable micro-actions. We discuss implications for HITL collaborative scaffolding under latency and state misalignment.

## CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; Accessibility systems and tools; Collaborative and social computing systems and tools; • **Computing methodologies** → Artificial intelligence.

## Keywords

Dementia care, older adults, human-AI collaboration, human-in-the-loop, voice scaffolding, vision-language model, text-to-speech, structured creativity, feasibility study

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## 1 Introduction

Dementia is an umbrella term for progressive conditions that impair memory, cognition, and everyday functioning beyond typical aging, often accompanied by changes in mood, behavior, and motivation. It remains incurable and primarily affects older adults. The World Health Organization (WHO) estimates that in 2021 over 57 million people lived with dementia worldwide, with nearly 10 million new



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cases each year; its impacts extend beyond individuals to caregivers, families, and society through substantial physical, psychological, social, and economic burdens [19].

Arts participation has been associated with reduced depression and agitation [17, 21], improved mood and engagement [12], and potential cognitive benefits among older adults with dementia [20]. From a non-pharmacological perspective, visual art therapy may alleviate depression and anxiety and support cognitive outcomes, where social interaction and reflective, creative processes may play an important role [15].

However, real-world care settings face strong heterogeneity in older adults' comprehension and behavioral responses, alongside limited facilitation resources, making "immediate, actionable guidance" a key challenge [5]. We therefore examine the feasibility of using generative AI and vision-language models (VLMs) as structured scaffolding: AI provides stepwise pacing and encouragement, while a human facilitator can translate abstract semantic prompts into concrete, actionable cues when needed.

Building on this motivation, we develop an AI-assisted drawing interface for people with cognitive impairment that uses a tablet and segmented voice guidance to scaffold structured creative practice. Here, collaboration refers to complementary roles in scaffolding interaction: AI provides pacing and structure, while humans translate semantics into executable action when breakdowns occur.

Our contributions are:

- We design and implement a low-cognitive-load drawing interface integrating a VLM with localized text-to-speech (TTS) to deliver canvas-aware, segmented voice scaffolding and error-tolerant encouragement.
- In a one-on-one feasibility study at a day-care center (n=6; CDR [6] 1–4), we find that only a subset of participants can complete the task independently with AI voice scaffolding; more often, a human facilitator is needed as a semantics-to-action translation layer, highlighting a core bottleneck and design opportunity in dementia-care technologies: converting abstract instructions into actionable guidance.

## 2 Related Work

### 2.1 Digital Creative Arts in Dementia Care

Non-pharmacological interventions are increasingly used to support the well-being of people living with dementia. Prior work has explored technology-supported creative activities for individuals with MCI or dementia, including music, art-making/appreciation, and interactive objects [14], and examined their impacts on flow and social engagement in art-therapy contexts [9]. Studies also suggest tablets are a feasible medium for art-making with people with dementia [18], and digital drawing tools may benefit family caregivers as well [10].

To reduce reliance on scarce professional facilitation, ePAD introduced AI-based automated prompts (visual/audio) triggered by engagement estimates and customizable by art therapists. While participants generally responded positively to tablet drawing, prompting effects were limited: prompts were infrequent, and even when

noticed, participants rarely responded as intended [11]. This highlights challenges in prompt salience and action alignment, suggesting that automation alone is insufficient and that more context-aware support is needed.

### 2.2 AI-Enabled Scaffolding for Creative Activities in Dementia Care

AI has been applied to dementia-related interventions (e.g., arts, horticulture, music, reminiscence, and emotional support) [9]. However, AI-and-drawing research in this area largely targets automated assessment, particularly Clock Drawing Test (CDT) analysis using deep models (e.g., CNN, ViT) to improve screening for MCI/dementia [1, 2]. These approaches treat drawing as a diagnostic instrument; in contrast, our work frames drawing as a creative activity and uses AI as process-oriented scaffolding for empowerment and co-creativity rather than pathological screening. Meanwhile, recent generative-AI tools for creators either decompose image generation into human-like stages for iterative discussion and refinement (e.g., SakugaFlow [8]) or provide critique based on computer-vision analysis of visual attributes (e.g., ArtKrit [13]); however, they are designed for general creators and rarely address dementia-related cognitive and fine-motor constraints. Overall, dementia-focused creative systems often use simple touch or non-interactive media and lack semantically grounded, context-specific prompts. VLM-based scaffolding may improve earlier prompting approaches (e.g., ePAD) by generating timely, actionable guidance aligned with the user's drawing state.

## 3 System Design

We developed an art-creation support system that integrates generative AI with a vision-language model (VLM). Through stepwise voice guidance and an animated companion, the system aims to help older adults with dementia engage in tablet-based creative activities despite cognitive and physical constraints (Fig. 2).

### 3.1 System Architecture

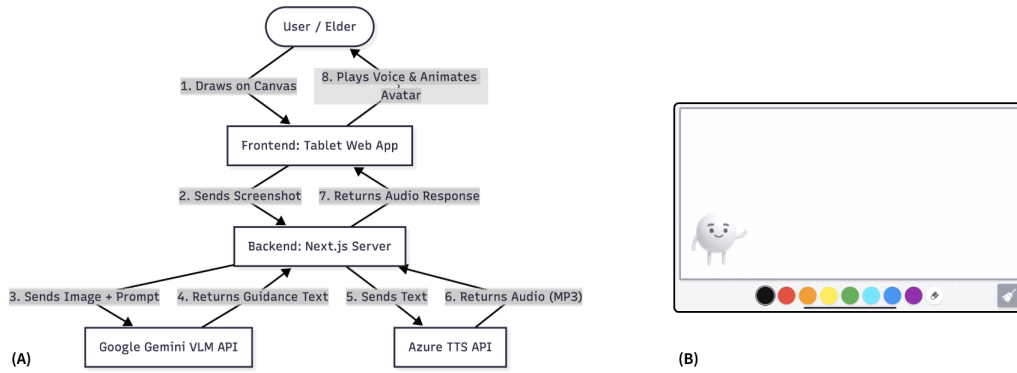
The system uses a client-server design for low-latency interaction. A Next.js tablet client captures strokes, renders the companion, and handles audio playback (stylus-first input). The server calls the Google Gemini VLM API to analyze the current canvas snapshot and drawing history, and uses Azure Text-to-Speech (TTS) to generate localized, moderately paced voice feedback.

### 3.2 User Interface Design

To accommodate vision decline, limited fine-motor control, and cognitive load [7, 16], the UI follows high-contrast, low-distraction principles (Fig. 2). The screen prioritizes a large canvas with a single row of eight high-contrast color buttons, an eraser, and a clear-canvas tool. A 2D companion in the lower-left corner provides a sense of presence and animates its mouth during speech.

### 3.3 Backend Logic Framework

The VLM is used to convert visual progress into instructional scaffolding. Inference is triggered when drawing pauses and passes



**Figure 2: System overview: (A) client-server pipeline for canvas capture, VLM reasoning, and localized TTS; (B) low-clutter tablet UI with large color buttons and an animated companion.**

validity checks (valid stroke; no in-flight request). The system summarizes what has been drawn and suggests the next step and/or a color change. Prompting constrains output to short, colloquial second-person utterances delivered in the voice of an art therapist using locally grounded language (e.g., “Tap green and add a leaf to this flower.”).

### 3.4 Voice Interaction Response Mechanism

To avoid speech conflicts during playback, we use a turn-taking strategy that **locks voice triggering during audio, marks new valid strokes as pending, and batches them into a single post-playback update** by capturing the latest canvas state to generate the next utterance (Fig. 3).

To maintain attention, the system replays the last guidance if inactivity exceeds 60 seconds.

## 4 User Study

We conducted a single-group feasibility study in a day-care center. All sessions were one-on-one to support close observation and capture individual experiences.

### 4.1 Participants & Ethical Considerations

We recruited 11 older adults through a local day-care center. Three participated in a pilot, and two were excluded due to physical limitations that prevented interaction. The core analysis included six participants with cognitive impairment ( $n=6$ ; CDR 1–4).

The study received IRB approval. Participants or legal proxies provided informed consent; when participants were judged unable to consent, guardians signed on their behalf. All data were de-identified, and sessions were stopped immediately if participants showed fatigue or refusal.

### 4.2 Study Design & Procedure

Each session lasted 20–50 minutes and comprised three phases:

**Phase 1: Warm-up (5–10 min).** The researcher built rapport and assessed readiness.

**Phase 2: Tutorial & AI-guided task (5–20 min).** Participants first practiced basic actions (color selection, drawing, erasing) for

5–10 minutes. In the main task ( $\leq 15$  min), participants completed a structured creativity task under AI voice scaffolding. The shared flower reference (petals, leaves, stem, pistil) was selected to ensure pragmatic observability for consistent coding of task progression, while allowing participants freedom in color, strokes, and composition. An element was coded completed if it was reliably identifiable from shape/placement or explicitly labeled by the participant during drawing; otherwise it was marked indeterminate. Completion and indeterminate labels were coded by one researcher via post-hoc video review. When needed, a human facilitator intervened by rephrasing instructions or using gestures.

**Phase 3: Post-study interview (10–20 min).** We conducted a semi-structured interview using a simplified, in-situ adaptation of SUS items [3], covering comprehension/affect, usability/preference, autonomy/willingness, and overall satisfaction (5-point Likert).

### 4.3 Preliminary Findings

We used a mixed-methods evaluation, combining quantitative metrics (task completion rate [4] and task time) with qualitative analysis from in-situ observation and video review. We examined older adults’ behavioral responses to AI voice scaffolding and how a human facilitator intervened at breakdowns by translating prompts into actionable micro-instructions; short post-task interviews captured subjective experience.

**4.3.1 Quantitative Overview: Task Completion & Time.** Task completion was defined over four sub-goals (petals, leaves, stem, pistil) and coded with two criteria: lenient (visually identifiable or explicitly labeled during drawing) and conservative (visually identifiable only; indeterminate counted as incomplete). With AI guidance and optional human assistance, 5/6 participants completed all four elements under the lenient criterion ( $\approx 83\%$ ); P05 diverged from the task goal. Under the conservative criterion, element-level completion was 17/24 ( $\approx 71\%$ ), with P10–pistil and P11–petal/pistil marked indeterminate. Task time (from onset of the first AI utterance) ranged 1:20–13:43 (median=6:46). Without human intervention, only 2/6 (P04, P08) completed the main elements under AI voice guidance (mean task time=1:24;  $n=2$ ). In this small sample, we found no clear linear trend between CDR and completion; observations instead

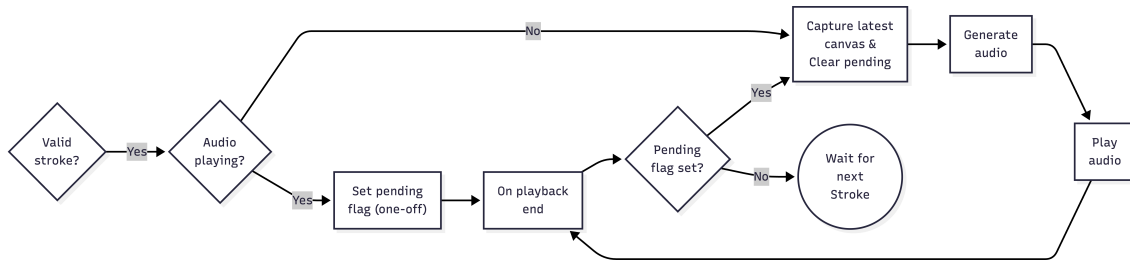


Figure 3: Overview of the Voice Feedback Mechanism.

suggested performance differences were related to participants' ability to comprehend and act on AI voice prompts.

#### 4.3.2 Qualitative Findings: Scaffolding Dependency, Instruction Comprehension, and Social Dynamics.

*Task pathways (scaffolding dependency).* We observed three patterns: **AI-independent (2/6)** participants treated the AI as an interaction partner and adjusted actions to prompts; **human-supported (3/6)** participants repeatedly required a human facilitator as a translation layer to turn abstract prompts into executable micro-actions (e.g., pointing/gestures), with AI speech mainly maintaining pace and direction; and **task divergence (1/6)**, where repetitive scribbling was difficult to redirect, suggesting voice scaffolding may not interrupt certain behavior loops.

*Conversational rhythm and state alignment.* Effective interactions relied on responsive, conversational pacing, but we also observed occasional state misalignment (repeated prompts after an element appeared completed), plausibly due to timing delays in screenshot updates or request queuing (not validated via logs).

*Semantics-to-action gap.* Progress depended on whether semantic understanding translated into action: participants responded more reliably to concrete cues (e.g., shapes, color keywords, deictic instructions) than to abstract labels (e.g., part names), often requiring operationalized prompts (e.g., "tap here").

*Subjective experience and social presence.* Interviews were broadly positive (satisfaction:  $M=3.83/5$ , median=4; enjoyment:  $M=4.17/5$ , median=4), yet enjoyment did not imply independence or reuse; some preferred having someone nearby, and one participant explicitly liked the animated companion.

## 5 Discussion

This feasibility study suggests that most participants could complete the structured creative goal when AI guidance and human assistance were both available, whereas only a small subset could do so without human intervention. We therefore frame the system as a human-in-the-loop (HITL) collaborative AI scaffolding tool rather than a labor-saving device. In this context, the system functions as a feasibility probe for 1:1 facilitated sessions, where "benefit" is defined by successful task engagement and creative agency rather than total independence. The AI provides pacing, encouragement, and next-step direction, while a human facilitator serves as a translation layer that converts abstract prompts into

actionable cues at breakdowns, consistent with prior evidence that automated prompting can fail when it does not translate into next actions [11].

A recurring challenge was a semantics-to-action gap: participants responded more reliably to concrete, executable cues (e.g., shapes, color keywords, deictic actions) than to abstract labels (e.g., "petal," "change color"). This implies that scaffolding should prioritize actionable micro-steps (concrete verbs + targets) and support lightweight facilitator intervention. Furthermore, we observed occasional state misalignment—such as repeated prompts for a completed element—likely due to processing latency or asynchronous canvas updates. While such misalignment may increase the facilitation burden by requiring human assistants to resolve the confusion, we observed that it did not necessarily discourage participants; some simply continued their own creative trajectory, often by adding more elements to the canvas.

To mitigate these issues, future iterations should incorporate designs such as clear state feedback or explicit confirmation steps to ensure the AI's internal state remains synchronized with the user's physical progress. Additionally, while the system is not a total labor-saving device, it may significantly reduce the cognitive "decision cost" for facilitators by providing a structured interaction script. This potentially allows a single caregiver to supervise multiple semi-autonomous stations in a day-care workflow, intervening only when the AI-human-user loop encounters a breakdown. Finally, positive experience did not necessarily imply independence or reuse, and some participants preferred a social context, suggesting the system may function as both a functional tool and a companionship-oriented activity within the broader ecology of dementia care.

### 5.1 Limitations

Completion was coded by a single researcher, so inter-rater reliability was not estimated; despite explicit rules and conservative handling of indeterminate cases, interpretation error is possible. This single-group feasibility study has a small  $N$  and heterogeneous CDR (1–4), so we do not claim causal effects or group differences. Furthermore, our findings regarding AI scaffolding effectiveness may be task-structure dependent, as the selected flower-drawing task provided a clear, stepwise progression for the VLM to monitor. Consequently, these results may not generalize to open-ended art therapy or unstructured creative activities without further qualification. We did not quantify end-to-end latency or verify delay sources via logs, and post-task interviews used an in-situ simplified

instrument adapted from SUS items [3], which may be affected by interviewer framing and momentary affect.

## 5.2 Future Work

Future work will quantify “valid stroke → speech onset” latency from system logs and relate it to pausing or distraction to tune concurrency control. We will also formalize the translation layer with one-click actionable rephrasings, stronger deictic supports (e.g., highlights/region cues), and a systematic comparison of prompt forms (geometry/color keywords vs. part semantics) to reduce the semantics-to-action gap while avoiding over-polishing that may diminish agency or ownership [22].

## 6 Conclusion

We presented an AI-guided drawing tool combining a VLM with localized voice feedback and evaluated it in a day-care feasibility study. Most participants completed the structured creative goal with AI plus human assistance, but only a few did so independently. Our findings highlight design opportunities in reducing the semantics-to-action gap, treating facilitator translation as a designable collaboration layer, and sustaining coherent interaction under latency and state misalignment.

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